

Node-RED:

3. Introduction to Node-RED & Dashboard

Scenario:

Cisco's Lab CL238 Temperature Monitoring Dashboard A user uses a Node-RED dashboard to set a simulated room temperature and observe how the temperature changes.

Dashboard Component

Slider

Gauge

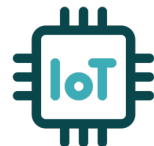
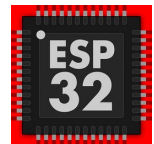
Chart/Graph

Function

Allows the user to set the temperature from 0°C to 50°C

Displays the current temperature

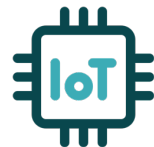
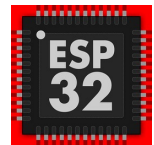
Shows the temperature changes over time



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STEPS:

1. Develop StoryBoard

2. Configure the properties first.

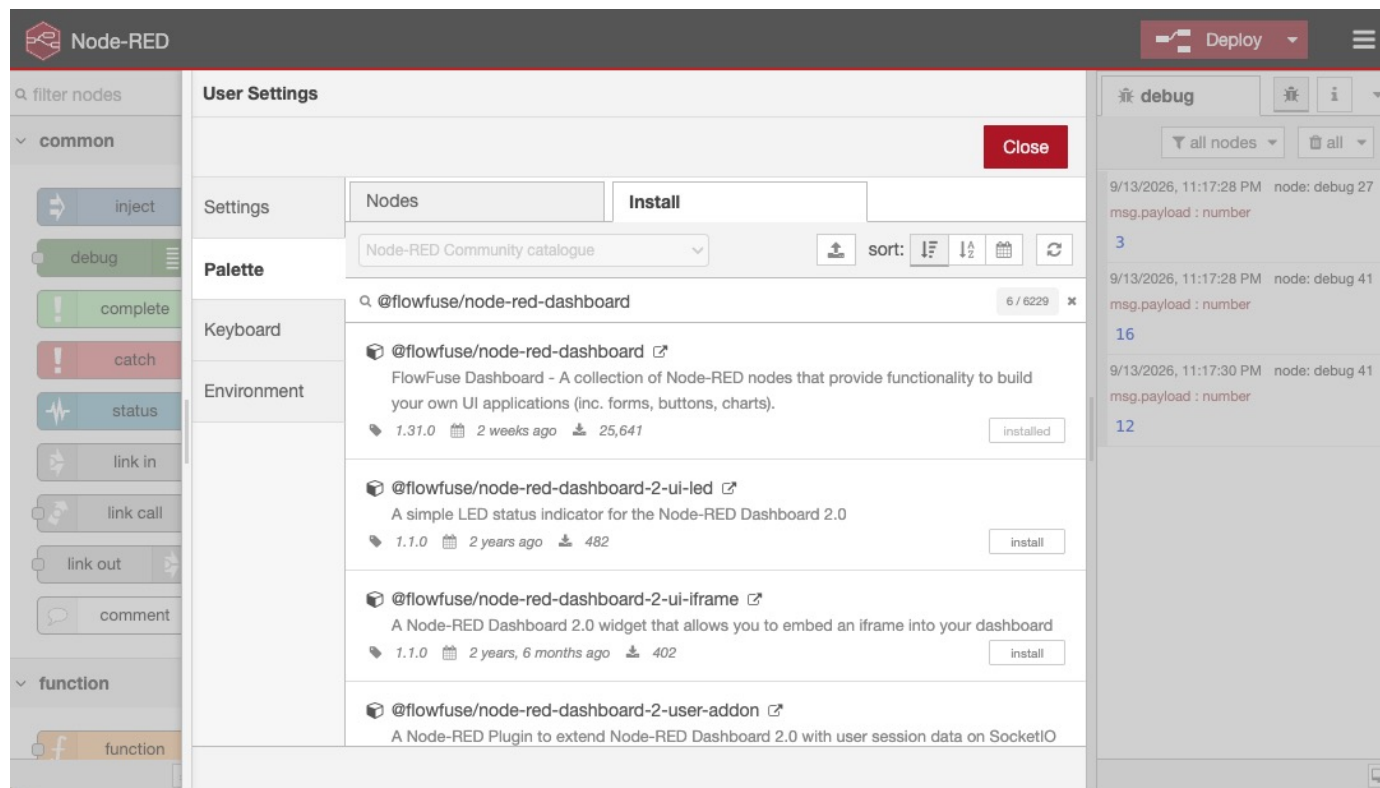
Sidebar Tab Menu > Dashboard 2.0 > Pages > Group > Widgets

3. Add nodes

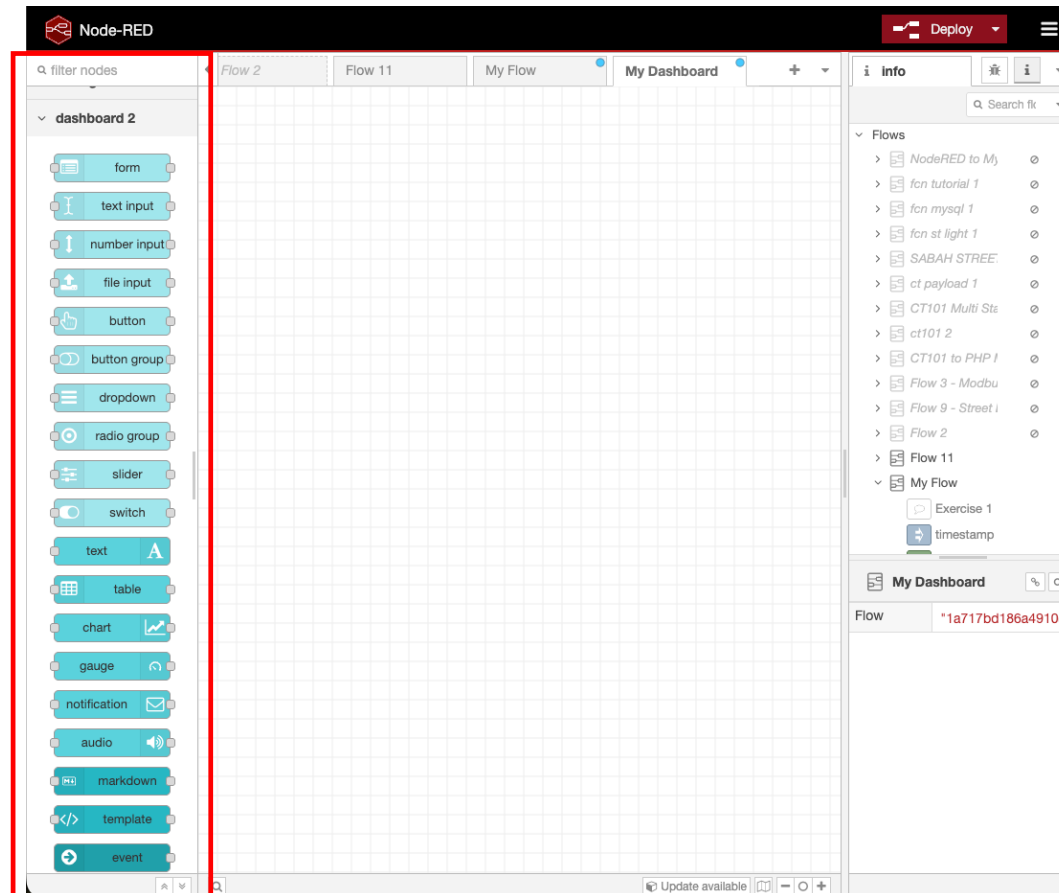
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i. Add Dashboard 2.0 palette

Menu >> Manage palette >> Install >> @flowfuse/node-red-dashboard >> Install >> Close >> Deploy

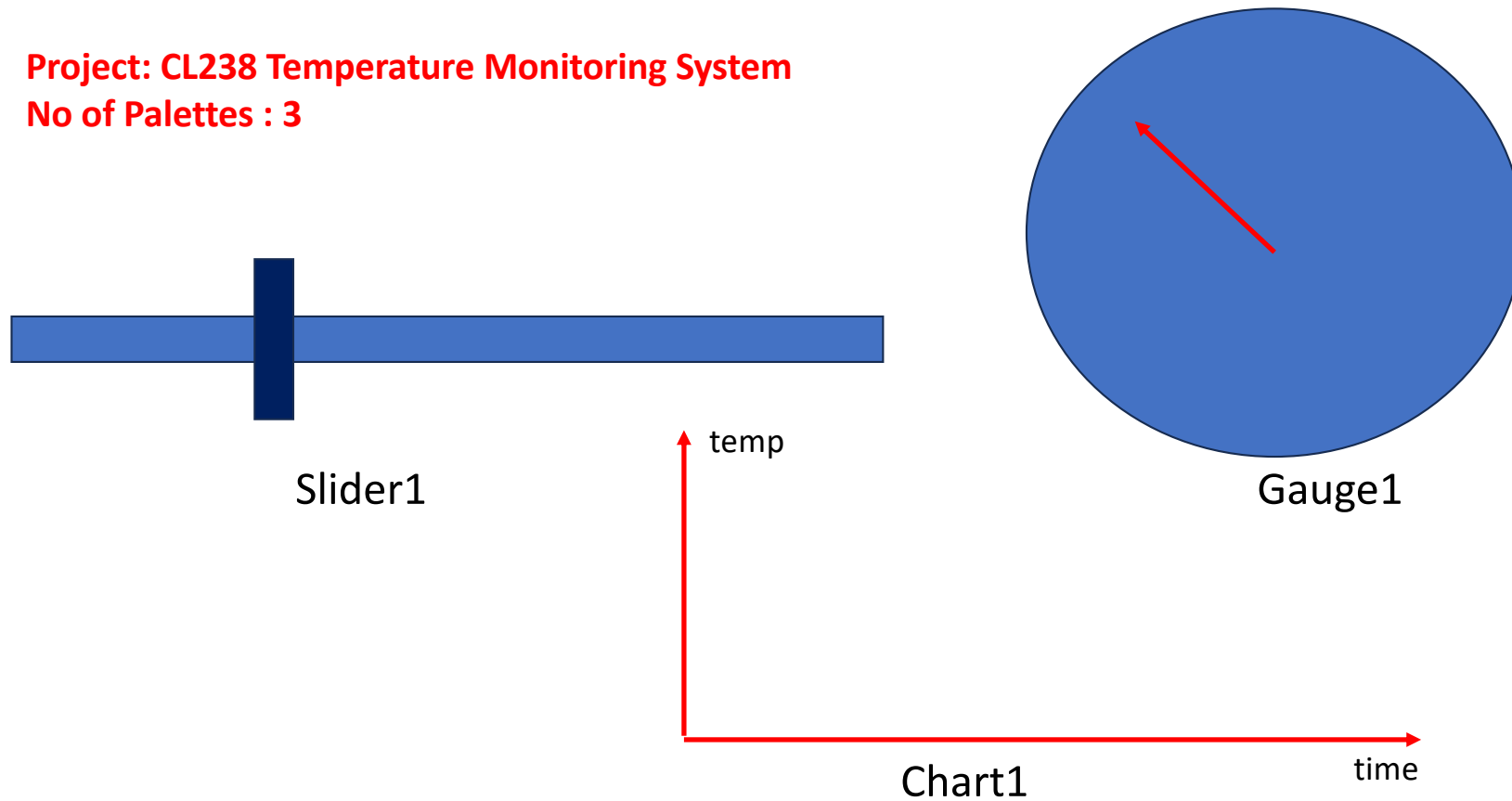


ii. Check the Dashboard 2 Palette



iii. Plan a Story Board

Project: CL238 Temperature Monitoring System
No of Palettes : 3



iv. Setup Dashboard Properties

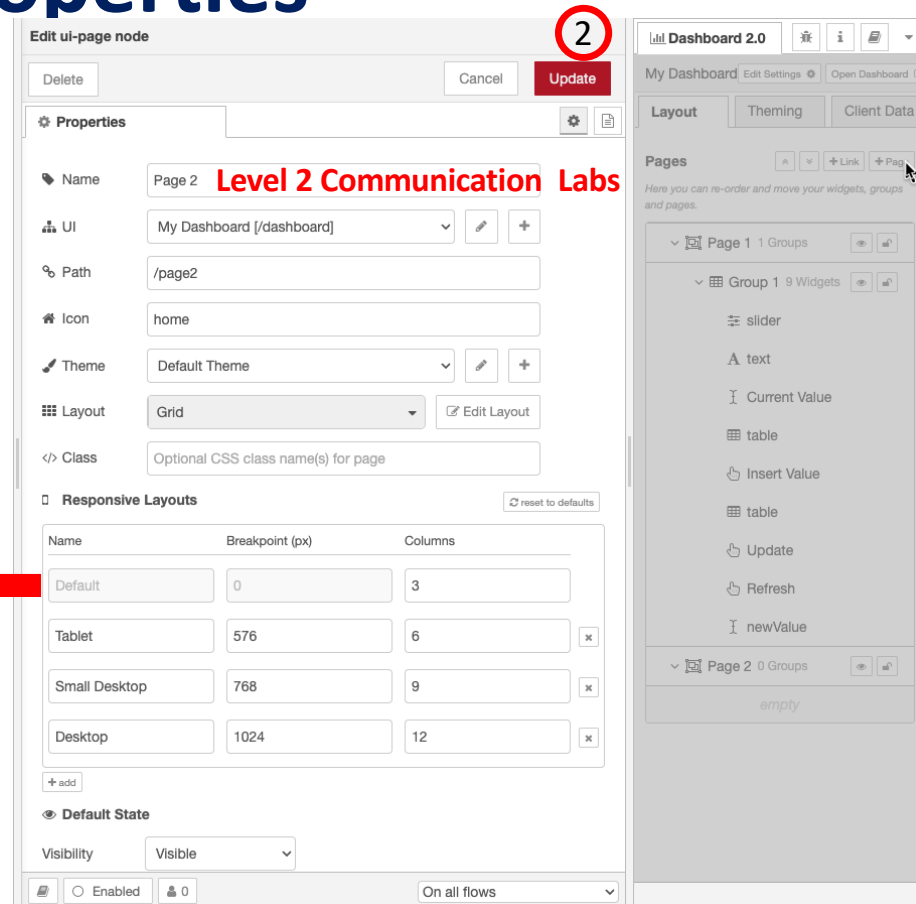
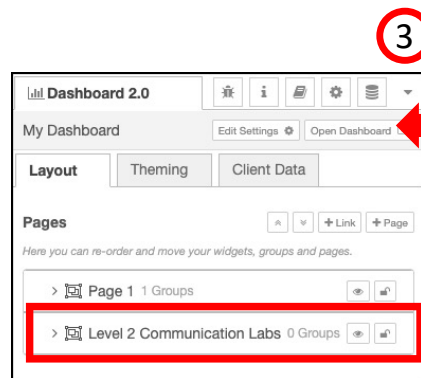
The screenshot shows the Node-RED web interface. On the left is the 'dashboard 2' sidebar with various widgets. The main workspace is empty. On the right, the 'debug' console shows a list of messages. A red box labeled '1' highlights the 'info' icon in the top right of the debug console. A red arrow points from this box to a second red box labeled '2', which highlights the 'Dashboard 2.0' option in the 'info' menu. Another red arrow points from box '2' to a third red box labeled '3', which highlights the '+ Page' button in the 'Dashboard 2.0' settings panel. The settings panel shows tabs for 'Layout', 'Theming', and 'Client Data', with the 'Pages' section expanded.

1

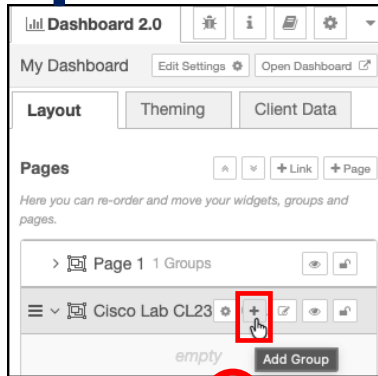
2

3

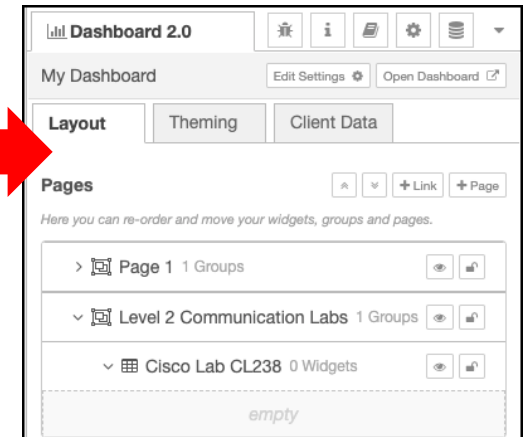
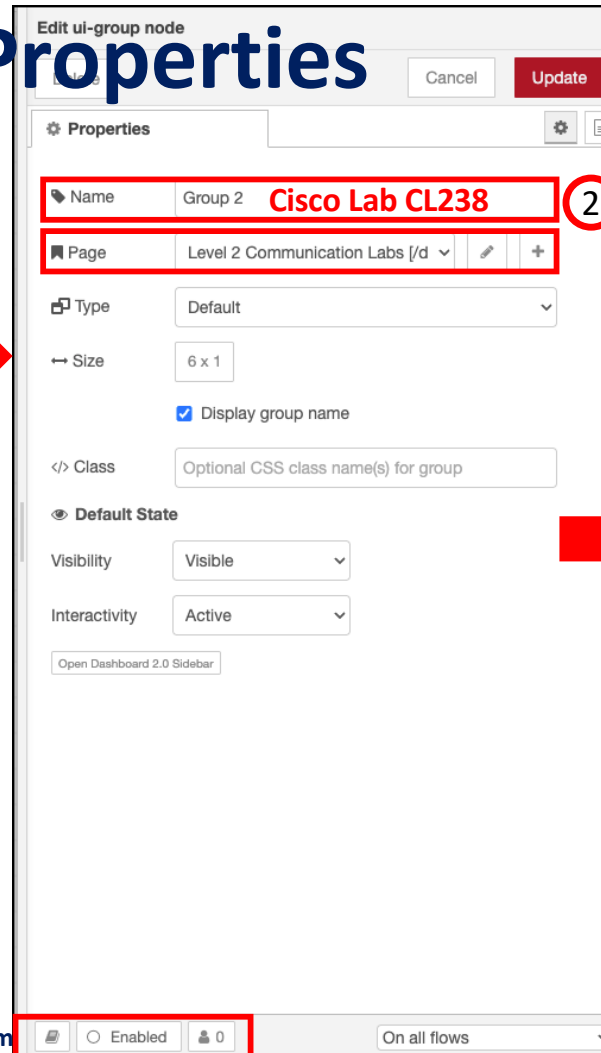
iv. Setup Dashboard Properties



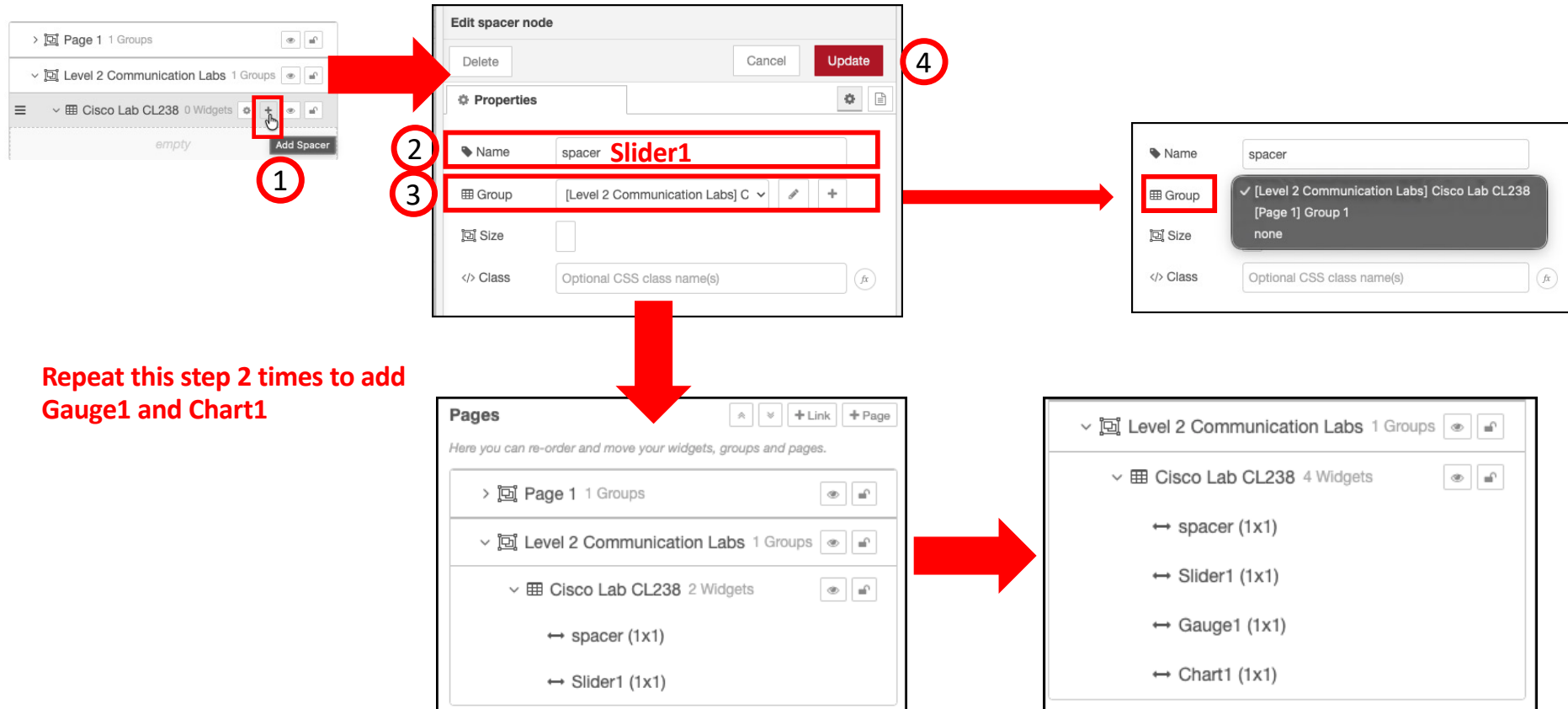
iv. Setup Dashboard Properties



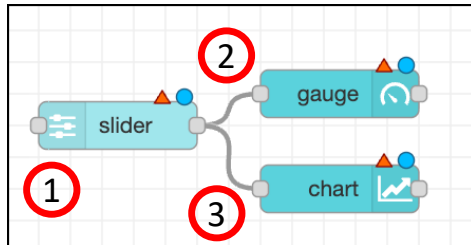
- Move your cursor at Cisco Lab CL238 tab, additional menu will appear
- Click + to add group



iv. Setup Dashboard Properties



v. Setup Dashboard Widgets



Edit slider node

Delete Cancel Done

Properties

Name

Group none

Size auto

Class Optional CSS class

Label slider

Text Field ☐ Show

Thumb On Drag

Ticks Always

Range min 0 step 1 max 10

Icons before minus after plus

Color slider red track green thumb #a5a5a

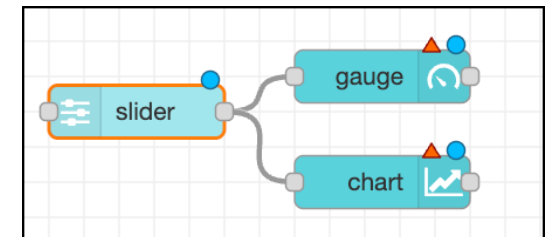
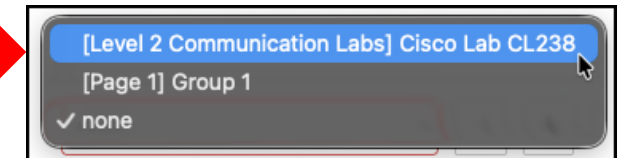
Output continuously while sliding

If msg arrives on input, pass through to output: ☐

When changed, send: ☒

Topic msg.topic

Enabled



Repeat this step 2 times to add Gauge1 and Chart1

v. Setup Dashboard Widgets

Edit gauge node

Delete Cancel Done

Properties

Name

Group [Level 2 Communication Labs] C

Size 3 x 3

Type Half Gauge

Style Needle

Value

Value msg.payload

Limits

Range min. 0 max. 10

Segments

0	Label
4	Label
7	Label

Defaults Enabled

Edit chart node

Delete Clear Chart History Cancel Done

Properties

Name

Group [Level 2 Communication Labs] Cisco Lab CL238

Size 6 x 8

Label chart

Class Optional CSS class name(s)

Chart Configuration

Type Line

Interpolation Linear

Action Append

Point Style Circle

Radius (px) 4

Axis Configuration

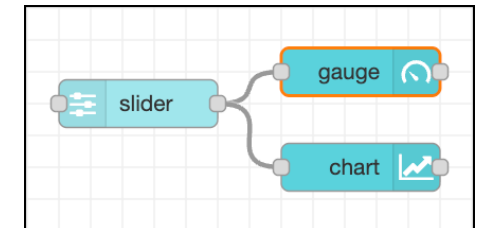
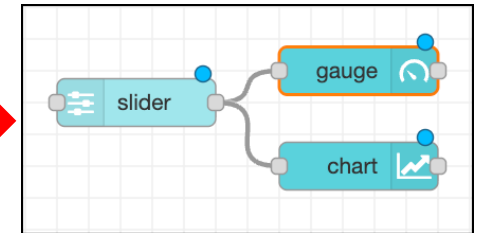
X-Axis Type Timescale

Format Automatic

X-Axis Label

X-Axis Limit last 1 Hours OR 1000 points

Enabled



vi. Dashboard Preview

Edit gauge node

Delete Cancel Done

Properties

Name

Group [Level 2 Communication Labs] C

Size 3 x 3

Type Half Gauge

Style Needle

Value

Value msg.payload

Limits

Range min. 0 max. 10

Segments

0	Label
4	Label
7	Label

Defaults Enabled

Edit chart node

Delete Clear Chart History Cancel Done

Properties

Name

Group [Level 2 Communication Labs] Cisco Lab CL238

Size 6 x 8

Label chart

Class Optional CSS class name(s)

Chart Configuration

Type Line Show Legend ☒

Interpolation Linear

Action Append

Point Style Circle Radius (px) 4

Axis Configuration

X-Axis Type Timescale

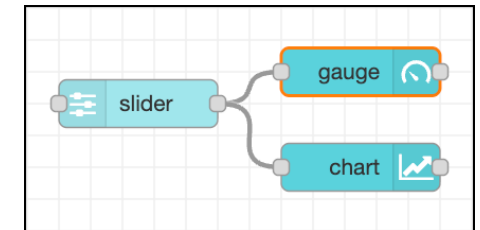
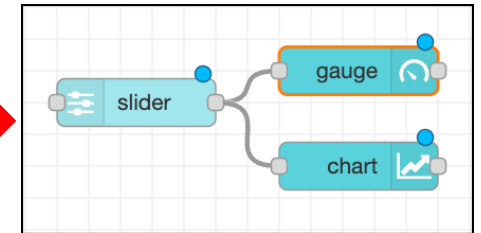
Format Automatic

X-Axis Label

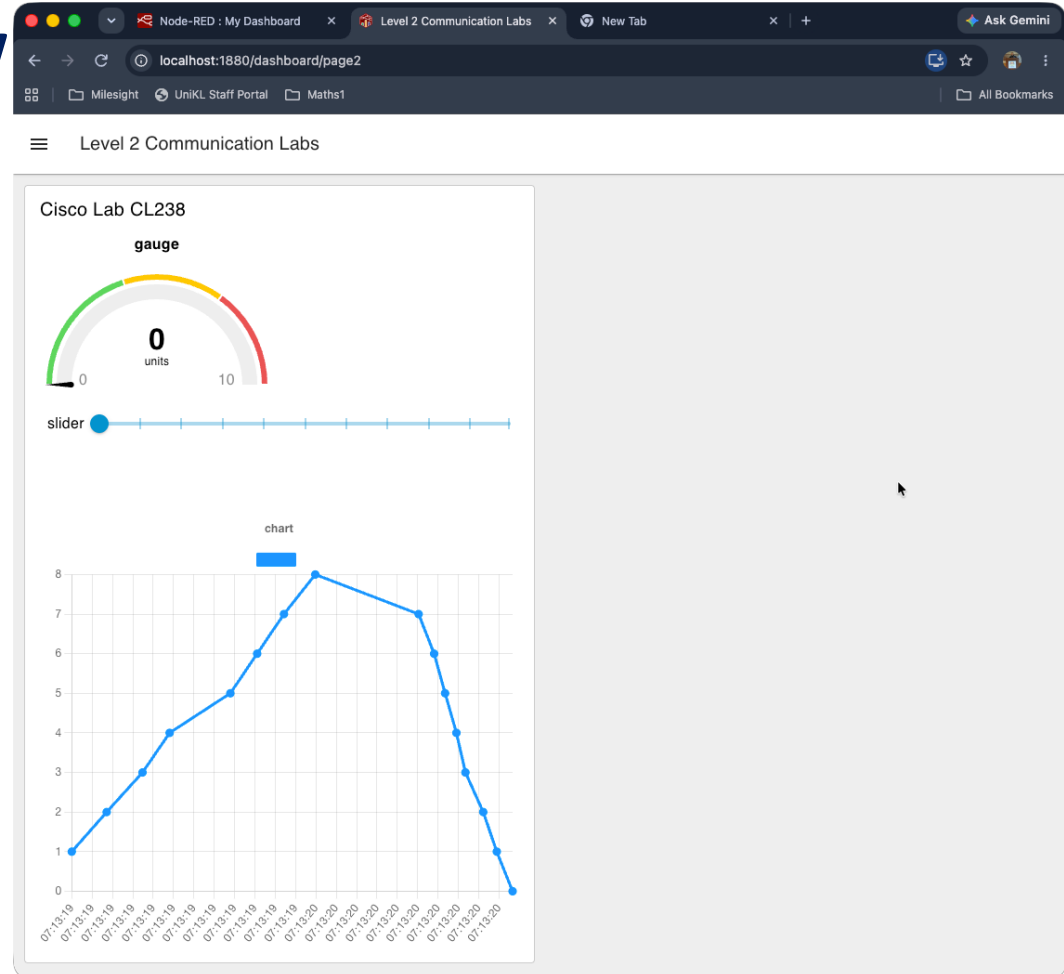
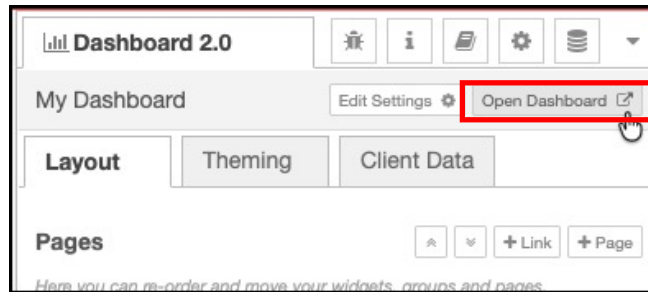
X-Axis Limit last 1 Hours OR 1000 points

X-Axis Label

Enabled

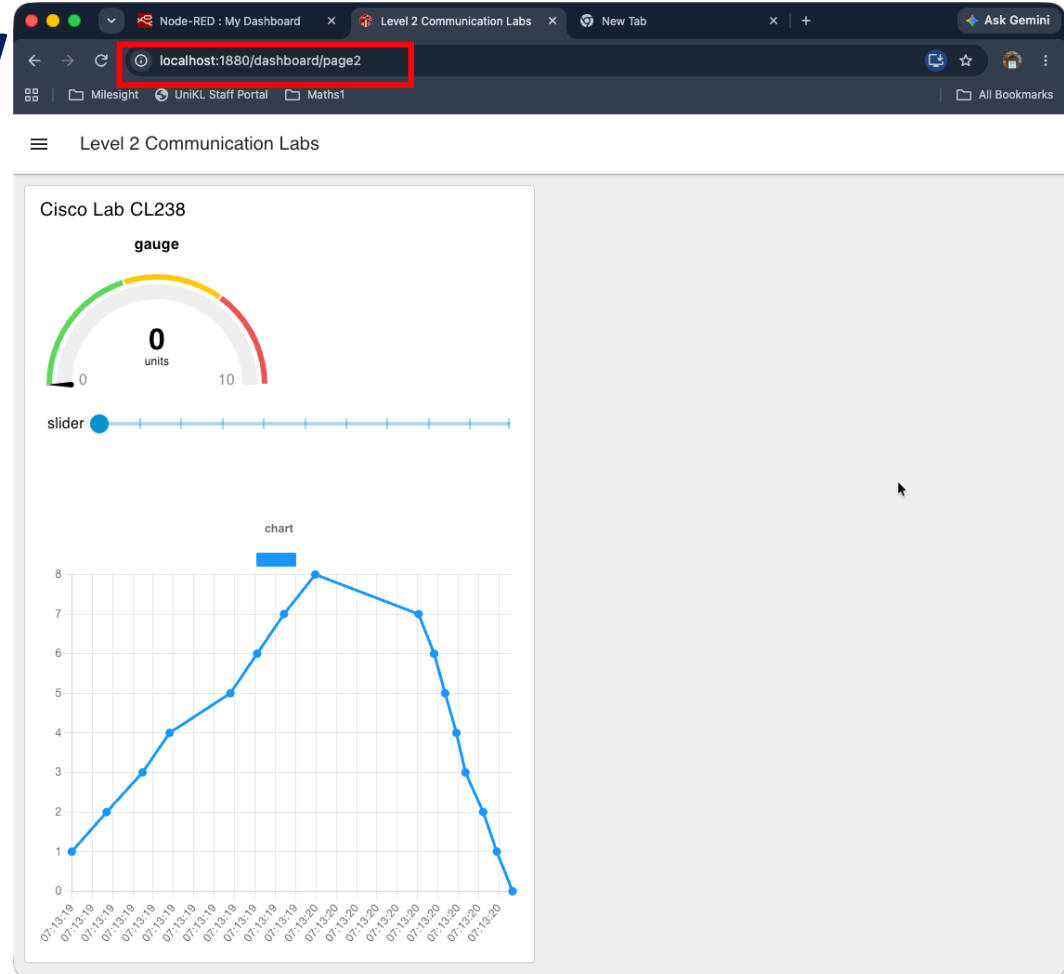


vi. Dashboard Preview



vi. Dashboard Preview

Go to widget setup properties and play around with the settings.



END